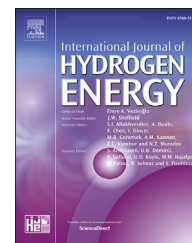


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A cold start mode of proton exchange membrane fuel cell based on current control

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HIGHLIGHTS

- A cold start model including temperature distribution and heat exchange is presented.
- Features between ice formation and thermal behavior are clarified by simulation.
- A current control strategy considering ice and heat absorption of membrane is raised.
- The novel mode improves start-up time and ice volume fraction at low temperatures.
- A method to avoid degradation by current regulating through inertia link is proposed.

ARTICLE INFO

Article history:

Received 9 June 2021

Received in revised form

2 October 2021

Accepted 17 November 2021

Available online 7 December 2021

Keywords:

PEMFC

Cold start

Fuzzy control

Heat transfer

Ice volume

ABSTRACT

In this study, a mathematical model is established to simulate the cold start of fuel cell, including the calculation of the temperature distribution and heat exchange. Moreover, a novel cold-start mode is designed and compared with the constant and linear current cold-start modes. It uses the ice volume and heat absorbed by the membrane as fuzzy control inputs and outputs current density. Compared with other modes at 263 K, the cold startup time is shortened by 25.6–41.6 s, and the ice volume fraction is reduced by 29.4%–31.8%. Only the proposed mode achieves a successful cold start at a lower temperature. Also, the proposed mode has better thermal behavior, as indicated by the temperature distribution diagrams. Furthermore, to avoid performance degradation caused by cold starts, an inertia link is added to the controller, so that the current amplitude is reduced by 7.98%, and the maximum change rate by 57.44%.

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Introduction

Proton exchange membrane fuel cells (PEMFCs) have the advantages of excellent power density, high efficiency, and low emissions. Accordingly, they have been developed and applied in various fields [1–3]. However, if the temperature of the external environment is below zero, the water produced by PEMFC is likely to form ice in the porous medium and cause startup failure; this has become one of the important factors

hindering the commercialization of PEMFC [4–6]. The cold start of PEMFC consists in starting the fuel cell from a sub-zero temperature and running to the operating temperature. During this period, the heat and ice generated by the fuel cell compete with each other [7]. If the fuel cell temperature fails to rise above the freezing point before the ice covers the reaction area, it will stop working and the cold start will fail. This will cause irreversible loss to the PEMFC components and performance [8–10], thus it is essential to fully understand the principles of cold start and design an effective start mode.

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<https://doi.org/10.1016/j.ijhydene.2021.11.139>

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Numerous studies have been concerned with the mechanism and characteristics of cold start through extensive simulations. Meng et al. [11] developed a transient multiphase multi-dimensional PEMFC model in a mixed-domain framework, whereby it was demonstrated that the water vapor concentration inside the cathode gas channel affected ice formation in the cathode catalyst layer. Yao et al. [12] investigated the freezing mechanisms of super-cooled water and ice formation using a transient three-dimensional multiphase model, and introduced a freezing probability function. Jiao et al. [13] observed that the ice in the cathode catalytic layer (CL) was formed faster at the section close to the bipolar plate (BP) at a lower current density, and it was close to the membrane at a higher current density. Gwak et al. [14] explored ice melting and transient water transport characteristics as well as the resultant dynamic responses under various operating conditions in terms of the air flow rate, external thermal boundary condition, inlet RH, and startup temperature. Ouyang et al. [15] proposed a dimension-reduction simplification method to obtain a model with a lower computational cost. Amamou et al. [16] experimentally demonstrated that the model proposed by Amphlett had higher accuracy than the Squadrito and Boulon models in terms of the prediction of characteristics under above and below zero temperature conditions. Khandelwal et al. [17] investigated the effectiveness of various warming strategies in reducing the start-up time by developing a one-dimensional thermal mathematical model. Zhou et al. [6] developed a one-dimensional cold-start model; furthermore, a novel start-up method, namely, variable heating and load control (VHLC), was proposed and evaluated based on this model. One-dimensional models are suitable for calculating the heat generation of a fuel cell and the mutual heat transfer of its internal structure. In particular, when the ice volume fraction is calculated without considering its specific distribution, a one-dimensional model can better balance the modeling details and calculation efficiency.

By considering various cold-start modes, several schemes have been proposed to optimize the cold-start process [18–25]. Du et al. [18] demonstrated that, compared with constant voltage and constant current modes, the maximum power cold start mode could better balance heat generation and ice, thus improving cold start survivability. Lin et al. [19] studied the cold start characteristics of the PEMFC stack in the constant voltage start mode. They found that the stack had better cold start performance than a single fuel cell and achieved a fast cold start within 95s at a temperature of -15°C . Luo et al. [20] clarified the fundamental difference between the constant power, constant current and constant voltage cold start modes, and observed that the heating effect had little bearing on the start mode, which was mainly determined by the current density. They recommended that maintaining a high current density during cold start was always beneficial to stack heating. Jiang et al. [24] used a lumped thermal model and proposed a current-ramping strategy for rapidly starting up PEMFCs from a subfreezing environment. It has been demonstrated that changing the cold-start mode such that a larger amount of heat and a lower ice volume fraction are generated can improve the start-up capability of the fuel cell. However, the minimum PEMFC cold start temperature in these modes is approximately -10°C . This is because the

relationship between heat generation and ice is ignored when the current density and voltage are changed. It is premised that the current density must be changed under full consideration of the thermal behavior so that the cold start mode can be better formulated.

Moxley et al. [26] presented the first quantitative analysis of ignition in PEMFCs and found their steady-state multiplicity. They cleverly attributed the main contradiction between water production and removal to the diversity problem. However, there is no further study on the cold start method and mode, and the effect of uneven internal temperature distribution is ignored.

In this study, a numerical model considering multiphase heat and mass transfer is established to simulate the cold start process. Through this model, we can calculate the ice volume fraction, temperature distribution, heat production, and heat exchange between the layers of the PEMFC. Furthermore, the present study uses ice volume and heat absorbed by the membrane as fuzzy control inputs, and outputs the current density so that a cold start mode is designed. Compared with other schemes, this cold start mode uses the influence of thermal behavior as the basis of the fuel cell output and provides new insights into the design of a cold start strategy. The cold start process of the PEMFC is analyzed according to the temperature distribution of the fuel cell under different ice volumes or membrane heat absorption. The numerical results clearly indicate the difference in thermal performance between the proposed cold start mode and other modes at low temperatures. The control effect of the inertia link is also investigated. The paper is structured as follows. Numerical mode aims at the numerical model, which forms the basis of the entire study. Results and discussion validates the model and proposes a new cold start mode. The simulation results of various modes are also compared. The major conclusions are stated in the last section.

Numerical model

Basic assumption

The establishment of the model is the key to the research on the cold start of PEMFC. The quantitative formulation of heat transfer and water transmission is fundamental and should be clarified first [15,17]. The following assumptions are made in this analysis:

- (1) Each layer is one-dimensional during the operation of the fuel cell, and its temperature is uniform. It can be assumed that the membrane electronics assembly (MEA), gas diffusion layer (GDL), BP, gas channel and cooling channel have uniform (but different) temperature distributions. The MEA is composed of the membrane (MEM) and CL.
- (2) The produced water can be frozen in the cathode CL and GDL, but only the ice in the CL of the cathode blocks the reaction area. Because the thickness of the CL is a few microns, the invalid reaction area caused by ice blocking can be expressed as the ice fraction s , whereas the effective area is $1-s$.

- (3) The components of the reactive substance in the cell and the flow channel are uniform, that is, the reactive gas does not produce a pressure drop in the flow channel, and the temperature change does not affect the gas quality. Thereby, the constant parameters, namely, the specific heat at a constant pressure, and the gas mass flow rate, can be obtained, allowing the reaction rate to be expressed in terms of the constant reactant-gas concentration.
- (4) There is no liquid water in the fuel cell below the freezing point.

Thermal analysis of cold-start processes

In the MEA, the change rate of water is equal to the difference between the water generation rate owing to the fuel cell reaction and water transport through the boundary. When the temperature of the fuel cell is below the freezing point, the produced water may be in the form of ionomer, gas phase and ice phase. And ice will appear if the gas phase is saturated. The cold start of a fuel cell can be divided into the following four stages through the water phase transition:

The first stage: The fuel cell starts up, and the cathode CL produces water that accumulates in the ionomer and gas phase.

The second stage: Once the water vapor in the CL reaches saturation, the water produced accumulates as ice in the CL [14,27].

The third stage: If the temperature of the PEMFC reaches the freezing point before the CL completely freezes, the ice accumulated in the CL will begin to melt.

The fourth stage: When all the ice accumulated in the CL has melted, the fuel cell temperature further rises to the normal operating temperature.

The water produced by the reaction in the second stage freezes into ice at sub-zero temperature and the increase in the ice volume fraction leads to a decrease in the electrochemically active area. The decrease in conductivity and the hindrance of gas transmission slow down the reaction rate, reduce heat generation and accelerate the accumulation of ice. Furthermore, as the ice volume fraction increases, the latent heat required for melting ice also increases, leading to a

longer duration of the third stage. To solve the contradiction between heat generation and ice volume, a method for controlling the current density through the heat absorbed by MEM and ice volume is proposed to raise the heat generation under the same ice volume and makes the fuel cell start successfully at a lower temperature.

Analytical model

In the analysis model of this paper, the performance output of the battery is calculated based on the Tafel equation and the transmission parameters in the cold start, rather than by solving the charge conservation equation [28]. And the design and operating parameters of the fuel cell are listed in Table 1.

The output voltage is calculated as follows:

$$V = E_{Nerst} - V_{act} - V_{ohmic} - V_{con}$$

The four terms on the right hand side are the Nernst voltage, activation overpotential, concentration overpotential and ohmic overpotential. The ideal open circuit voltage of the fuel cell is also called the Nernst voltage. The calculation method is as follows:

$$E_{Nerst} = 1.23 - (8.5 \times 10^{-4})(T_{mea} - 298.15) + (4.3 \times 10^{-5})T_{mem}[\ln(p_{H_2}) + 0.5 \ln(p_{O_2})]$$

The activation overpotential is mainly due to overcoming the activation energy of the reaction when electrons pass through the MEM.

$$V_{act} = -\frac{RT_{mea}}{\alpha F} \ln \left(\frac{I_{ref}}{(1 - s_{ice} - s_{iq})^{0.5} j_0 c_{cl}} \right),$$

$$c = \frac{1 + \left(1 - \frac{1}{\epsilon}\right) \frac{0.21 p_c}{RT_{mea}}}{2}$$

Reactants cannot reach the reaction interface in time, resulting in an insufficient electrochemical reaction of the gas. The calculation method of voltage loss is as follows [29,30]:

$$V_{conc} = \frac{RT_{mea}}{\alpha F} \ln \left(1 - \frac{I}{j_D} \right), j_D = \frac{4Fc}{\frac{\delta_{cl}}{D_{eff}} + \frac{0.5\delta_{cl}}{D_{cl}^{eff}}}$$

Table 1 – Cell design and operating parameters [4,12].

Parameter	Value	Unit
Active area for reaction	235×10^{-4}	m ²
Thicknesses of MEM/CL/GDL	0.178/0.01/0.2	mm
Channel length/width/depth/rib width	100/1/1/1	mm
Contact angles in pore CLs/GDLs θ	110/110	°
Intrinsic permeability of CLs/GDLs K_0	$1.0 \times 10^{-13}/1.0 \times 10^{-12}$	m ²
Equivalent weight of dry MEM EW	1.1	kg mol ⁻¹
Densities of MEM/CL/GDL/BP/ice ρ	1980/1000/1000/1000/920	kg m ⁻³
Specific heat capacities of MEM/CL/GDL/BP/ice C	833/2683/568/1580/2050	J kg ⁻¹ K ⁻¹
Thermal conductivities of MEM/CL/GDL/BP k	0.95/1.2/1.5/20.0	Wm ⁻¹ K ⁻¹
Porosities of CL/GDL ϵ	0.5/0.6	—
Transfer coefficient of anode/cathode α	0.5/0.5	—
Stoichiometry ratio of anode/cathode ξ	2.0/2.0	—
Initial ice volume fraction	0	—

The ohmic overpotential reflects the impedance effect in the process of proton and electron transfer. According to Ohm's law, the calculation method of ohmic overpotential is as follows:

$$V_{ohmic} = V_{ohmic}^{elec} + V_{ohmic}^{proton} = -I(R^{elec} + R^{proton}), R^{proton} = \frac{r_{mem}\delta_{mem}}{A}$$

$$r_{mem} = \frac{181.6 \left[1 + 0.03 \left(\frac{I}{A} \right) + 0.062 \left(\frac{T_{mea}}{303} \right)^2 \left(\frac{I}{A} \right)^{2.5} \right]}{\left[\lambda - 0.634 - 3 \left(\frac{I}{A} \right) \right] \exp \left(4.18 \frac{T_{mea} - 303}{T_{mea}} \right)}$$

The produced water can diffuse into the MEM through the ionomer and evaporate into the GDL and flow channel. When the water generation rate of the fuel cell exceeds the rate of water leaving the CL and the gas water vapor is saturated, the ice accumulation rate is as follows [31]:

$$\dot{n}_{acc}^{H_2O} = \dot{n}_{cl}^{H_2O} - \dot{n}_{mem}^{H_2O} - \dot{n}_{gdl}^{H_2O}$$

The water yield of the oxygen reduction reaction, the actual water absorption of the MEM, and the water vapor transported from the CL to the GDL are calculated as follows:

$$\dot{n}_{cl}^{H_2O} = \frac{IA}{2F} (1 + n_d)$$

$$\dot{n}_{mem}^{H_2O} = \frac{\rho_{dry}(\lambda_{sat} - \lambda_0)}{EW} \delta_{mem} A$$

$$\dot{n}_{gdl}^{H_2O} = \dot{n}_{outflow}^{H_2O} = \zeta_c \frac{IA}{4F} \frac{1}{0.21} \frac{p_{sat}^{H_2O}}{p_c}$$

In the first stage, the water accumulated in the CL is absorbed by the ionomer, and the following equation can be obtained:

$$\int_0^{t_0} \dot{n}_{acc}^{H_2O} dt = \frac{\rho_{dry}(\lambda_{sat} - \lambda_0)}{EW} \epsilon \delta_{cl} A$$

In the second stage, the ice volume fraction is calculated as:

$$s = s_0 + \int_{t_0}^t \frac{\dot{n}_{acc}^{H_2O} M_{ice}}{\epsilon_{cl} \delta_{cl} A} dt = s_0 + \int_{t_0}^t \frac{(\dot{n}_{cl}^{H_2O} - \dot{n}_{mem}^{H_2O} - \dot{n}_{gdl}^{H_2O}) M_{ice}}{\epsilon_{cl} \delta_{cl} A} dt$$

The ice begins to melt in the third stage, and the ice volume fraction is calculated as follows:

$$s = s_1 + \int_{t_1}^t \frac{\left[\dot{Q}_{total} - (\dot{Q}_{a,gdl,bp} + \dot{Q}_{c,gdl,bp} + \dot{Q}_{a,gdl,gas} + \dot{Q}_{c,gdl,gas}) \right] M_{ice}}{\epsilon_{cl} V_{cl} h_{fusion}} dt$$

One-dimensional heat transfer model

The heat transfer process of the PEMFC is shown in Fig. 1, which mainly includes the heat generated by the MEA and the mutual heat transfer between the layers in the PEMFC. The heat generation in the MEM and CL is mainly reversible heat generated by the chemical reaction, irreversible heat caused

by the polarization phenomenon and latent heat of the water phase transition. The expression is as follows:

$$\dot{Q}_{generate} = \dot{Q}_{rev} + \dot{Q}_{irrev} + \dot{Q}_{sg}$$

$$\dot{Q}_{rev} = 9.0 \times 10^{-4} (T_{mea} - 298.15) I$$

$$\dot{Q}_{sg} = \frac{h_{sg} I}{2F}$$

$$\dot{Q}_{irrev} = (V_{act} + V_{conc} + V_{ohmic}) I$$

The numerical relationship of heat transfer between the layers is as follows [32]:

MEA Layer to GDL Layer:

$$\dot{Q}_{i,loss} = h_{mem,gdl} A_i (T_{mem} - T_{i,gdl}), i = anode, cathode$$

MEA temperature:

$$C_{mea} m_{mea} dT_{mea} = \dot{Q}_{generate} - \dot{Q}_{loss}$$

GDL Layer to GAS Layer:

$$\dot{Q}_{i,gdl,gas} = h_{gdl,gas} A_{gdl} (T_{i,gdl} - T_{i,gas})$$

GDL Layer to BP Layer:

$$\dot{Q}_{i,gdl,bp} = h_{gdl,bp} A_{gdl,bp} (T_{i,gdl} - T_{i,bp})$$

GDL temperature:

$$C_{gdl} m_{gdl} dT_{i,gdl} = \dot{Q}_{i,loss} - \dot{Q}_{i,gdl,gas} - \dot{Q}_{i,gdl,bp}$$

GAS Layer to BP Layer:

$$\dot{Q}_{i,gas,bp} = h_{gas,bp} A_{bp} (T_{i,gas} - T_{i,bp})$$

GAS Layer to gas:

$$\dot{Q}_{i,gas,out} = (\dot{m}C)_{i,gas,out} T_{i,gas,out}$$

GAS temperature:

$$C_{i,gas} m_{i,gas} dT_{i,gas} = \dot{Q}_{i,gdl,gas} - \dot{Q}_{i,gas,bp} - \dot{Q}_{i,gas,out}$$

BP Layer to CLANT Layer:

$$\dot{Q}_{i,bp,clant} = h_{i,bp,clant} A_{bp} (T_{i,bp} - T_{i,clant})$$

BP temperature:

$$C_{i,bp} m_{i,bp} dT_{i,bp} = \dot{Q}_{i,gdl,bp} - \dot{Q}_{i,bp,clant}$$

CLANT temperature:

$$C_{i,clant} m_{i,clant} dT_{i,clant} = \dot{Q}_{i,bp,clant}$$

Results and discussion

Comparison between model predictions and experimental data

Fig. 2 shows the comparison between the simulated and experimental voltage curves during two cold start processes at 253 K and 270 K respectively. The operating conditions and

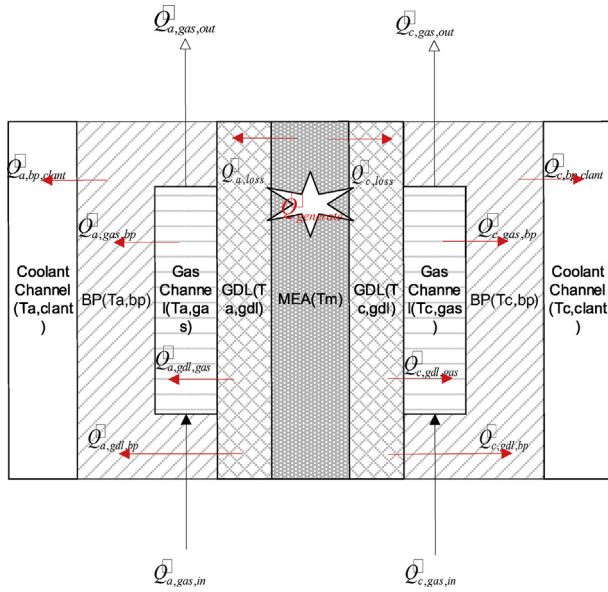
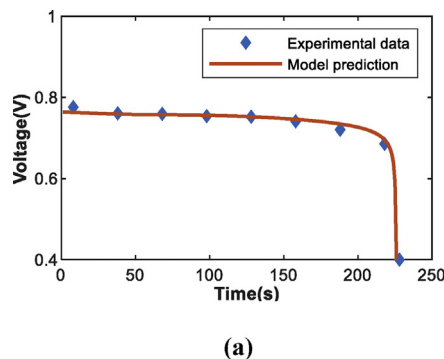


Fig. 1 – Heat transfer model of PEMFC.

PEMFC characteristics in the model are changed to be consistent with the experiment in Ref. [33] rather than the values in Table 1. The stoichiometric ratio is 2, the MEM thickness is 30 μm , the initial water content of the MEM and CL is 6.2, the initial ice volume fraction of the fuel cell is 0, and the relative humidity of the inlet air is 0. The current is linearly increased from 0 to 400 A/m^2 in 80 s, and then it stabilizes at that value. In this study, the model predictions obtained by starting fuel cell directly at a constant current density of 400 A/m^2 are compared with the experimental data after 40s (the time coordinates in Fig. 2 are 40s less than the measured values in Ref. [33]). In Refs. [4,27,34], the same method was used to compare the predicted results with the experimental data. As shown in Fig. 2, the measured voltage for the two cold start processes is higher in the first 40s because the experimental current increases linearly from 200 A/m^2 to 400 A/m^2 . The data predicted by the model are very close to the experimental data after 40s. It should be noted that the ice volume is linearly related to the reaction area without considering the influence of ice distribution. In general, the prediction accuracy of the model can meet the requirements of the variable-current cold start mode.



Cold start modes

Based on the above model, this study designs the cold start mode of the fuel cell and simulates eight different start conditions. The ambient temperature in Cases 1–5 is 263 K, and in Cases 6–8, it is at 253 K.

Cases 1 and 2 use constant current density of 400 A/m^2 and 500 A/m^2 throughout the cold start. In Case 3, the current density increases linearly from 400 A/m^2 to 500 A/m^2 within 100s. Case 4 linearly reduces the current density from 500 A/m^2 to 400 A/m^2 within 100s. In Case 6, the current density linearly rises from 900 A/m^2 to 1000 A/m^2 within 100s. In Case 7, the current density decreases linearly from 1000 A/m^2 to 900 A/m^2 within 100s. In Cases 5 and 8, the current variation is affected by the ice volume and heat absorbed by the MEM during the cold start. This relationship is explained in the next section.

Based on the previous analysis, a new start-up mode is designed to improve the operation of the fuel cell in low-temperature environment. The control scheme uses ice volume, heat absorbed by the membrane and reference current density as input, and outputs the corrected current density. Owing to the large inertia and nonlinearity of the fuel cell, it is very difficult to achieve the desired control effect using a conventional control strategy. Fuzzy control does not depend on the system model and is suitable for nonlinear systems. Therefore, a fuzzy control strategy for the cold start mode is developed. Considering the system response delay, the output current of the fuzzy control returns to the input of the fuel cell model after passing through the first-order inertia link.

In Cases 5 and 8, a fuzzy controller [35] is adopted to change the output current according to the ice volume fraction and heat absorbed by the MEM. More detailed information regarding the fuzzy controller is shown in Fig. 3. It consists of three main parts [36]:

1. The fuzzification interface uses fuzzy sets to convert specific inputs into fuzzy values.
2. The rule base and inference system generate a result for each appropriate rule, and then merge these results.
3. The defuzzification interface converts the combined result into a specific control output value.

Fuzzy reasoning has two inputs s and $\dot{Q}_{\text{mem}}$, and an output I . In Table 2, the fuzzy language variables are NB, NM, NS, ZO,

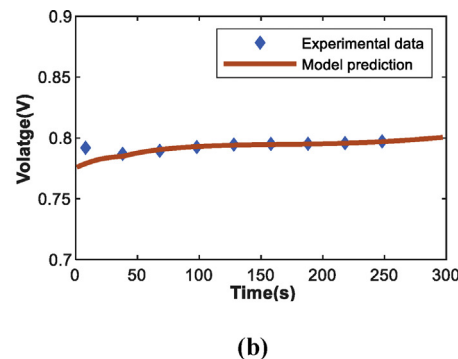


Fig. 2 – Voltage curves of model and experimental data. (a): at 253 K. (b): at 270 K.

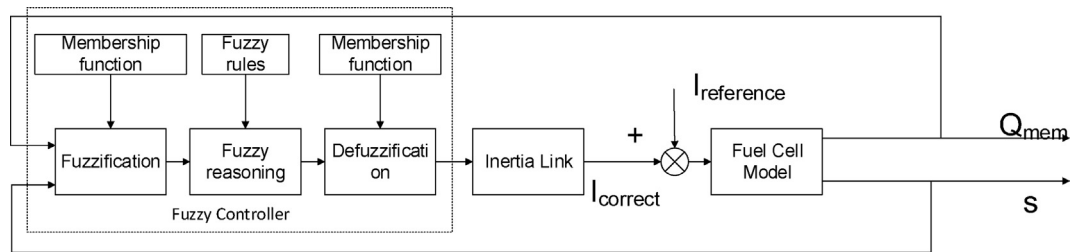


Fig. 3 – Flow chart of fuzzy control.

Table 2 – Fuzzy rules.

I		Ice Volume Fraction			
		ZO	PS	PM	PB
Q	NM	PB	PB	PB	PB
	NS	PS	PM	PM	PB
	ZO	PS	PS	PM	PM
	PS	ZO	ZO	PS	PM
	PM	NS	NS	ZO	ZO
	PB	NB	NM	NM	NS

PS, PM, and PB, which stand for negative big, negative medium, negative small, zero, positive small, positive medium, and positive big, respectively [37]. The basic form of the fuzzy control rule is: "If the ice volume s is A and $\dot{Q}_{\text{mem}}$ is B, then the fuzzy control output is I". The membership functions of the input and the output are shown in Fig. 4a, Fig. 4b, c. A surface diagram of the fuzzy control is shown in Fig. 4d. It can be seen that the output current reaches a maximum when the ice volume is large and the heat absorbed by the MEM is small; the

output current reaches its minimum value when the ice volume is small and the heat absorbed by the MEM is large.

It is worth noting that when the ambient temperature is 253 K, it is difficult to achieve a cold start by relying only on the heat generated by the fuel cell itself. Therefore, the output of the fuel cell is used to heat the heating wires on the BP for cold start in Cases 6–8 [38]. However, the current is still adjusted by fuzzy control without considering the influence of the heating wires, which heats the fuel cell at a constant voltage input of 0.4 V. To simplify the modeling and the influence of heat transfer efficiency, it is assumed that the heating effect is 0.3 V of the output of the fuel cell, which is directly converted into MEM heating power.

Cold start process at 263 K

Fig. 5 shows the time-dependent variation of current and voltage in Cases 1–5 at 263 K, whereas Fig. 5b shows the variation of the ice volume fraction and heat generation with the temperature of the MEM in Cases 1–5 at 263 K.

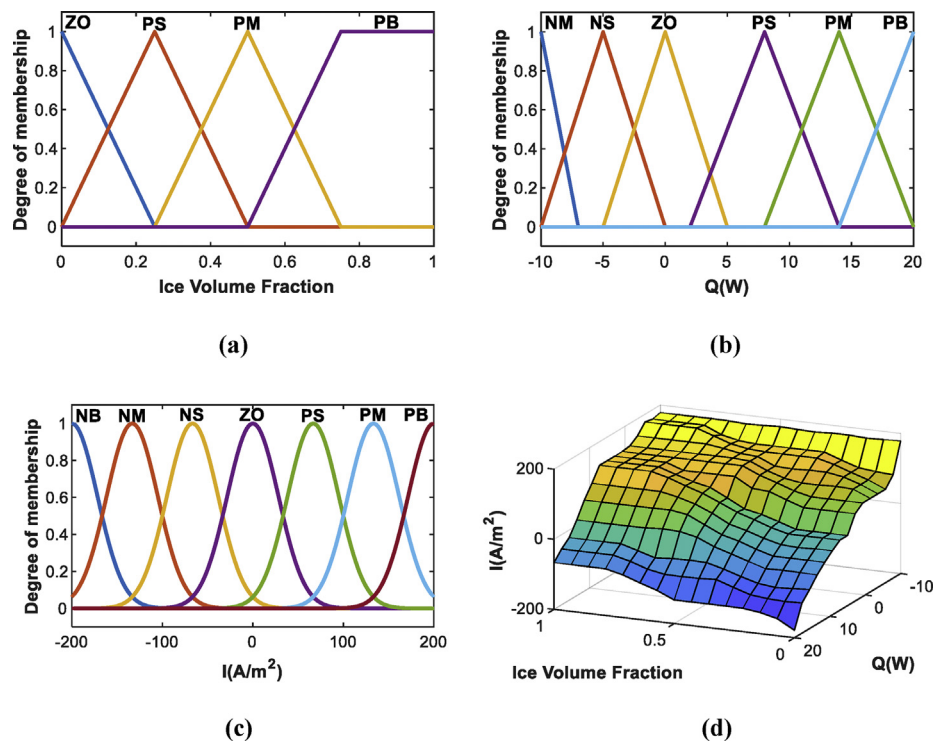


Fig. 4 – Membership function diagram. (a): Ice volume fraction (b): $\dot{Q}_{\text{mem}}$ (c): Current (d) Input and output diagram of fuzzy control.

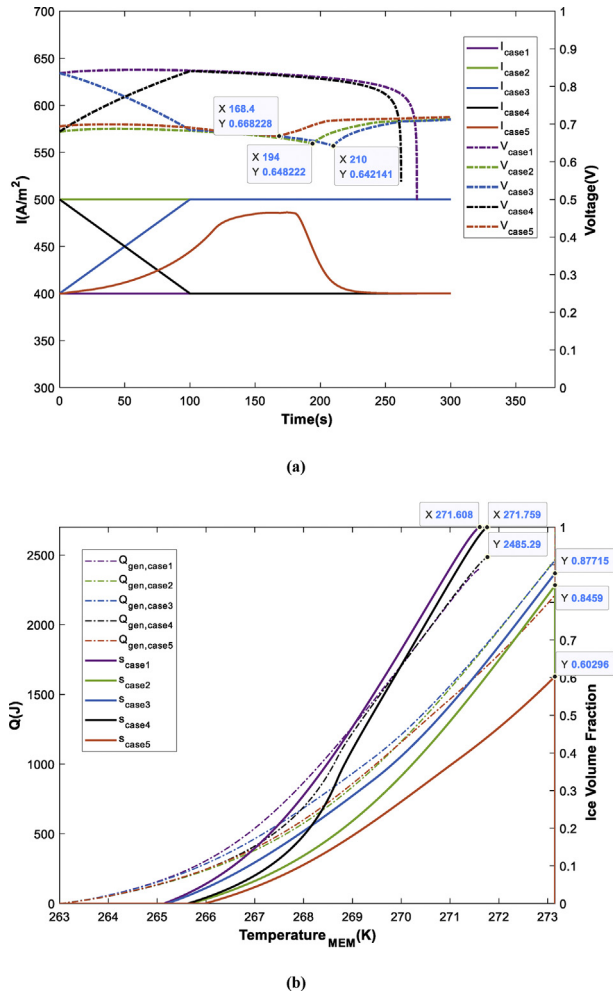


Fig. 5 – Cold start process at 263 K.(a)Current and voltage variation in Cases 1–5.(b)Ice volume fraction and heat generation in Cases 1–5.

The voltage in Cases 2 and 5 are essentially stable at approximately 0.68 V during the ice accumulation period, whereas the voltages of Cases 3 and 4 increase or decrease between 0.64 V and 0.84 V as the current varies. In Cases 1 and 4, the voltages rise slowly at the beginning owing to the effect of membrane hydration. The voltages go lower during the middle period, reflecting the change in ohmic resistance caused by the electro-osmotic drag effect and temperature increase. At 262s and 271s, the ice volume fractions in Cases 1 and 4 reach 1, and the voltages drop sharply from 0.8 V to 0 V because the ice fully covers the CL. At 194s, 210s and 168.4s, the maximum ice volume fraction is reached in Cases 2, 3, and 5, and the corresponding minimum voltages are 0.65 V, 0.64 V and 0.67 V. Subsequently, the voltages of Cases 2, 3 and 5 gradually increase to 0.72 V with ice melting. Compared with the time required for the voltage to stabilize in Cases 2 and 3, Case 5 has the fastest cold start.

When the MEM temperature rises to 273.15 K, the maximum ice volume fractions in Cases 2, 3 and 5 reach 0.85, 0.88 and 0.60, respectively. In Cases 1 and 4, the maximum MEM temperatures are 271.61 K and 271.76 K because of the increase in the ice volume fraction to 1. Interestingly, when

the temperature rises to the freezing point or the ice volume reaches 1, the maximum difference in heat generation is only 274 J, of which the lowest value is 2212.61 J in Case 5, and the highest is 2485.29 J in Case 4. The main reason is that when the current density in Case 4 goes down, the heat generation decreases, which is not enough to meet the melting heat corresponding to ice generation. And the low temperature rise rate extends the cold start time and leads to the increase of the heat generation required for unit temperature rise. In view of this, the heat absorption curve of the MEM and the curve of heat transfer to the GDL are provided for further study.

It can be seen from Fig. 6 that when the heat absorption curve is above the diagonal, the MEM absorbs sufficient heat before the ice volume fraction reaches 1 to raise the temperature to the freezing point, indicating a successful cold start. When the heat absorption curve is below the diagonal, the heat absorbed by the MEM is not enough to raise the temperature to the freezing point, indicating a failed cold start. From the curves for Cases 2 and 5, it can be seen that the MEM heat for rising from 263 K to the freezing point is approximately 332.98 J, which is closely related to the MEM mass and thermal physical parameters of MEM, as well as ambient temperature. The heat absorbed by the GDL in Case 5 is 1879.63 J and that absorbed by the GDL in Case 2 is 2132.99 J. This large difference may be due to the slow rise of MEM temperature in Case 2, resulting in a longer cold start time, and inducing the MEM to transfer more heat to the GDL. In Cases 1 and 4, the heat absorbed by the MEM are only 282.46 J and 287.64 J, whereas the heat absorbed by the GDL are 2118.62 J and 2197.66 J. It can be seen that in Cases 1 and 4, the heat absorbed by the MEM is lower than that in the other three cases, but the heat transferred by the MEM is higher. This also explains the failure of the cold start in Cases 1 and 4 when the MEM heat generation in these five cases is similar. Therefore, making the MEM absorb more heat and reduce heat transfer at the same ice volume fraction is the key to the success of cold start-up. The proposed mode has a lower current density than in Case 2 and 3 (slower ice formation) and higher heat absorption rate than in Case 1 and 4, thus achieving a better balance between heat generation and ice volume, and resulting in the best thermal behavior.

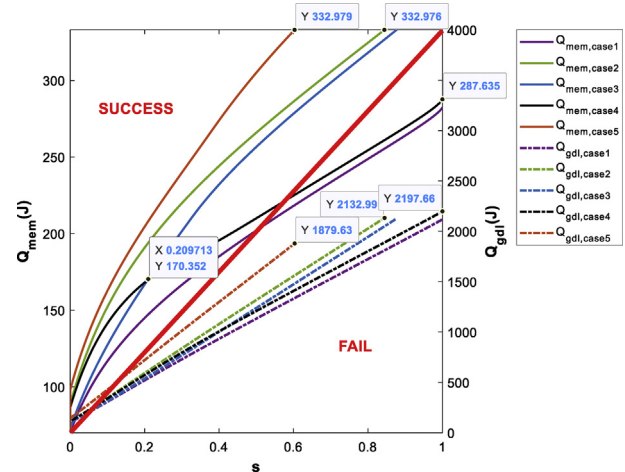


Fig. 6 – The heat absorbed by the MEM and transferred to the GDL in Cases 1–5 at 263 K.

It is worth mentioning that the ice volume fraction in Case 3 and Case 4 is only 0.33 at 100s, but the MEM heat absorption is equal when the ice volume is 0.21. Subsequently, the temperature in Case 3 is always higher, which is conducive to the successful cold start. This is because of the rapid temperature rise in the fuel cell owing to the high current, leading to a high level of heat exchange between the MEM and other layers. Therefore, to ensure a successful cold start, it should be increased from low current to high current as far as possible, rather than from high current to low current.

Fig. 7 shows the temperature distribution of the fuel cell when the MEM absorbs 200 J and 282 J (the maximum heat absorbed by the MEM in Case 1).

In Case 1 and Case 4, when the MEM absorbs two different amounts of heat, the temperature distribution difference of each layer is less than 0.1 K, which also occurs in Case 2 and

Case 3. Therefore, Case 1 and Case 2 are replaced by Case 4 and Case 3 respectively, and the temperature distribution of each layer is not shown in Fig. 7. The MEM absorptions in Case 1 and Case 4 are 200 J at 151.8s and 125.6s, and 282 J at 273.9s and 256.6s, confirming that the time difference has no effect on temperature distribution. The heat generation and internal heat transfer of the fuel cell are the unique properties of the system, which are only affected by the internal structure, material properties and external load current. Compared with the temperature distribution with respect to time, the plotted temperature distribution can better explain the internal heat transfer of the fuel cell when the MEM absorbs a different amount of heat. Controlling the current density is conducive to improving the temperature rising speed of the MEM and reducing the heat exchange with other structures.

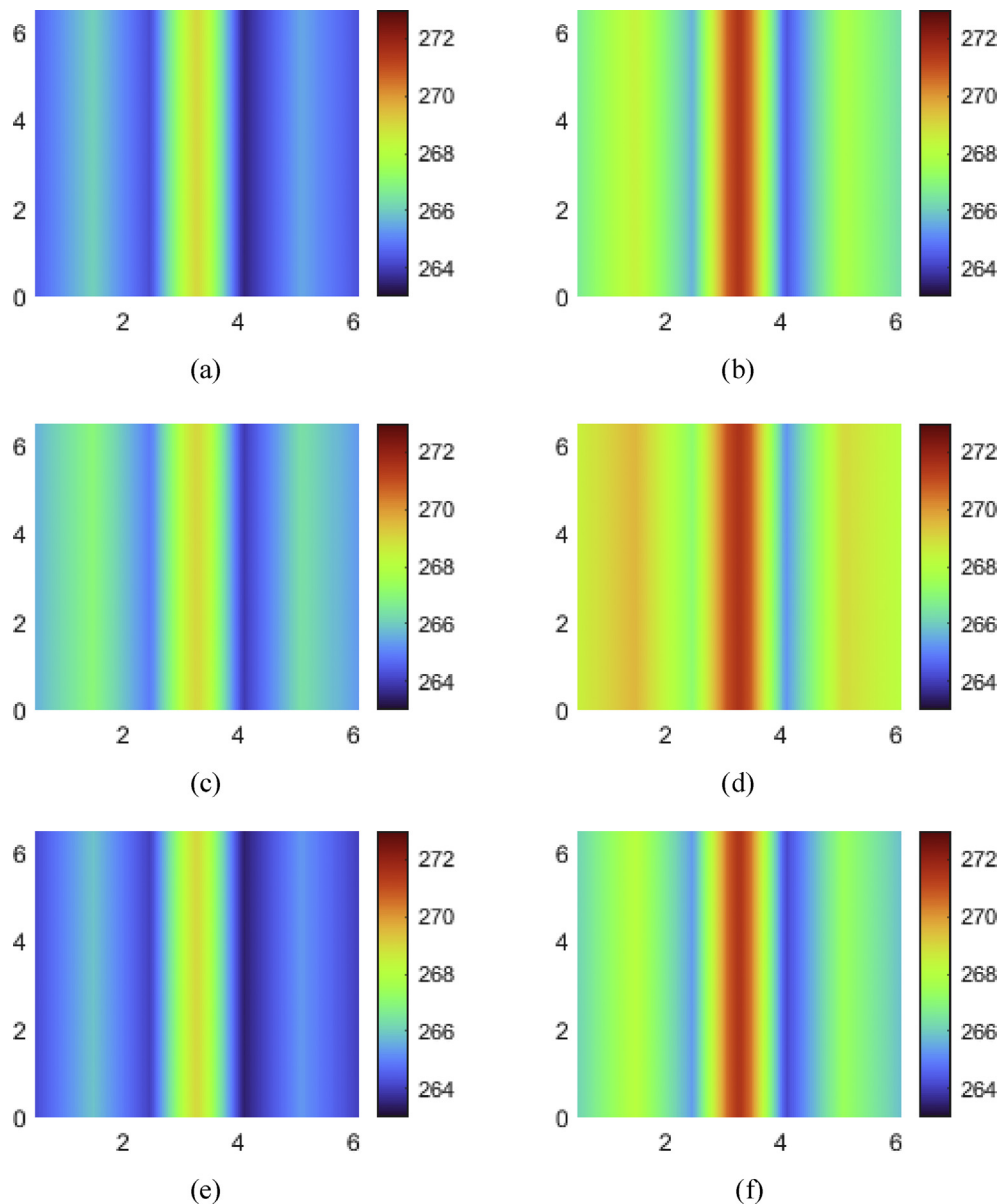


Fig. 7 – The temperature distribution of the fuel cell. (When MEM absorbs 200 J:(a) Case 3 (c) Case 4 (e) Cases 5; When MEM absorbs 282 J:(b) Case 3 (d) Case 4 (f) Cases 5).

As can be seen from Fig. 7 (a), (c) and (e), the temperature distributions in Case 3 and Case 5, in which cold start is achieved, are generally the same. And the temperature in Case 3 is higher than that in Case 5, in which the temperature difference of the GDL is less than 0.1 K, and the temperature differences of the gas channel, BP and coolant channel are 0.3 K. The temperature of the GDL in Case 4 is 0.3 K higher than that in Case 5, and the temperatures of the gas channel, BP and coolant channel are 0.9K–1.4 K higher than that in Case 5. It can be seen from Fig. 7 (b), (d) and (f) that the temperature distributions in Case 3 and 5 change little, except that the temperature differences of the gas channel, BP and coolant channel rise to 0.5 K. The temperatures of the gas channel, BP and coolant channel in Case 4 are 1.8K–2.4 K higher than that in Case 5. Therefore, the external structure in Case 4 has the highest temperature and absorbs most of the heat generated by the fuel cell, whereas the MEM has the lowest heat absorption rate (11.6%), leading to cold-start failure. The heat absorption rate of the MEM in case 3 is 15.7%. In Case 5, when the heat absorbed by the MEM is the same, the heat transferred to the external structure is the lowest, and the heat absorption rate of the MEM is the highest at 17.1%. The

simulation results indicate that when the internal layer of the fuel cell reaches the same temperature (the MEM absorbs the same amount of heat), the heat exchange with the outer layers in Case 5 is the least. The reduction in the total heat generation also facilitates the cold start easier.

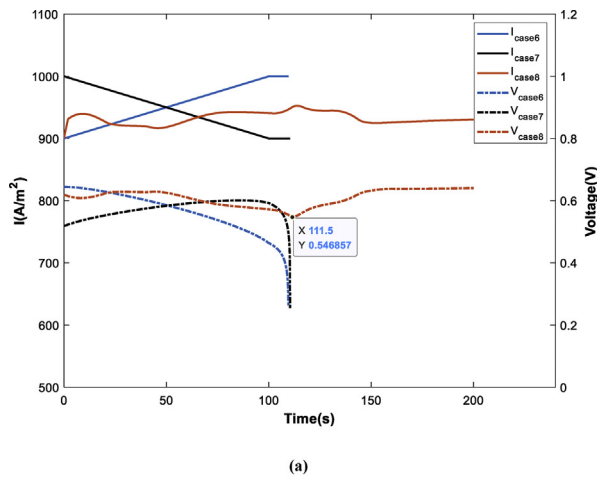
In the fuzzy-control-current mode, a fast cold start can be achieved by adjusting the current, and the largest possible amount of the heat generated by the fuel cell itself is converted into heat absorbed by the MEM, which is crucial for a successful cold start. As seen in Fig. 7, the temperature of the cathode side is slightly higher than that of the anode side. This is primarily because the specific heat capacity of the reaction gas hydrogen at the anode side is much higher than that of the air, directly causing the temperature of the anode GDL and BP to be lower than that of the cathode, and indirectly causing the temperature of the anode coolant channel to be lower than that of the cathode side through the BP temperature.

Cold start process at 253 K

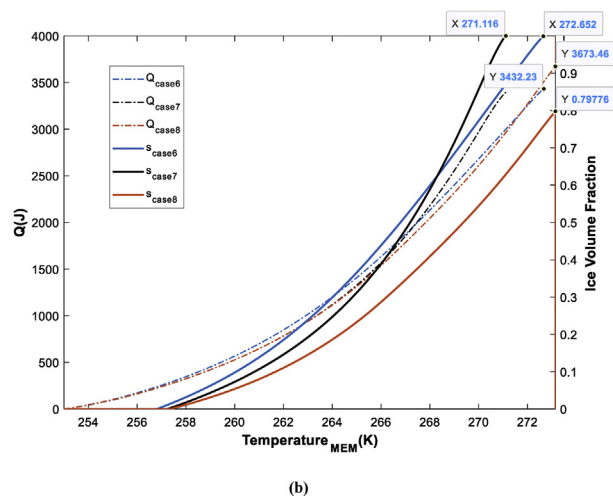
Fig. 8 (a) shows the time-dependent variation of the current and voltage in Cases 6–8 at 253 K; Fig. 8(b) shows the variation of the ice volume fraction and heat generation with respect to the MEM temperature in Cases 6–8 at 253 K.

During ice accumulation, the voltage in Case 8 is basically stable at 0.6 V, whereas the voltages in Cases 6 and 7 increase or decrease between 0.51 V and 0.64 V as the current changes. At 109.6s and 110.4s, the ice volume fractions in Cases 6 and 7 reach 1 and the voltages drop to 0 V. At 111.5s, the maximum ice volume is reached in Case 8, and the minimum value of the corresponding voltage is 0.55 V. Thereafter, the voltage in Case 8 gradually increases to 0.64 V. Compared with the simulation results when the ambient temperature is 263 K, only the proposed mode achieves successful cold start at 253 K.

When the MEM temperature rises to 273.15 K, the ice volume fraction in Case 8 reaches the maximum value of 0.80. In Cases 6 and 7, the maximum MEM temperatures are 272.65 K and 271.12 K, respectively, owing to the increase in the ice volume fraction to 1. In these three cases, when the temperature rises to the freezing point or the ice volume fraction



(a)



(b)

Fig. 8 – Cold start process at 253 K (a):Current and voltage variation in Cases 6–8. (b)Ice volume fraction and heat generation in Cases 6–8.

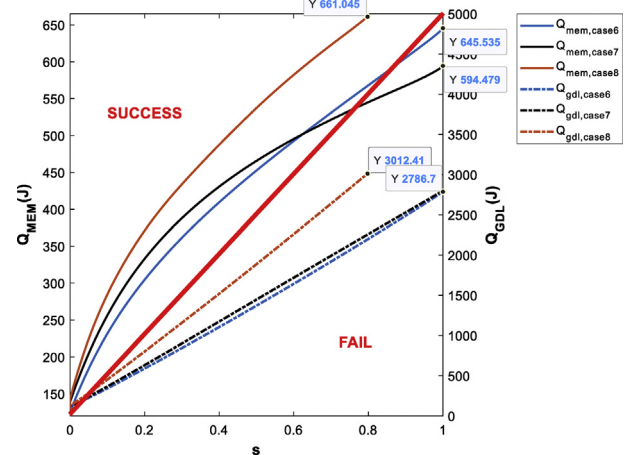


Fig. 9 – The heat absorbed by the MEM and transferred to the GDL in Cases 6–8 at 253 K.

reaches 1, the maximum difference of heat generation is only 241.23 J, of which the lowest value is 3394.91 J in Case 7 and the highest is 3673.46 J in Case 8. The maximum difference is smaller than that at 263 K, but they are approximately the same. Moreover, the heat generation is 1182.3 J higher than that at 263 K, which is primarily due to the lower ambient temperature and the increase in current density. In view of this, the heat absorption curve of the MEM and the heat transfer to the GDL are provided for further investigation.

According to the diagonal of the coordinate axis in Fig. 9, only in Case 8 is a successful cold start achieved. From the curve for Case 8, it can be seen that the MEM heat rising from 253 K to the freezing point is 661.05 J, and the heat absorbed by the GDL is 3012.41 J. In Cases 6 and 7, when the ice volume fractions reach 1, the heat absorbed by the MEM are only

645.54 J and 594.48 J, respectively, whereas the heat absorbed by the GDL are 2786.7 J and 2800.43 J. Compared with the ambient temperature at 263 K, the heat absorbed by the MEM should be increased to achieve a successful cold start at a lower ambient temperature, which also increases heat exchange with the other layers.

When the ice volume fraction is 0.80, the heat absorbed by the MEM and transferred to the GDL in Case 8 are nearly 20%–33% higher than in Cases 6 and 7 with the same ice volume fraction, and 2.5%–11.2% higher than in Cases 6 and 7 with ice volume fraction of 1. This not only indicates that in Case 8, the most heat is produced and the fastest, but also explains why only in Case 8 is a successful cold start achieved under the boundary conditions, even if the maximum heat generation difference is only 241.23 J in these three cases. This is different

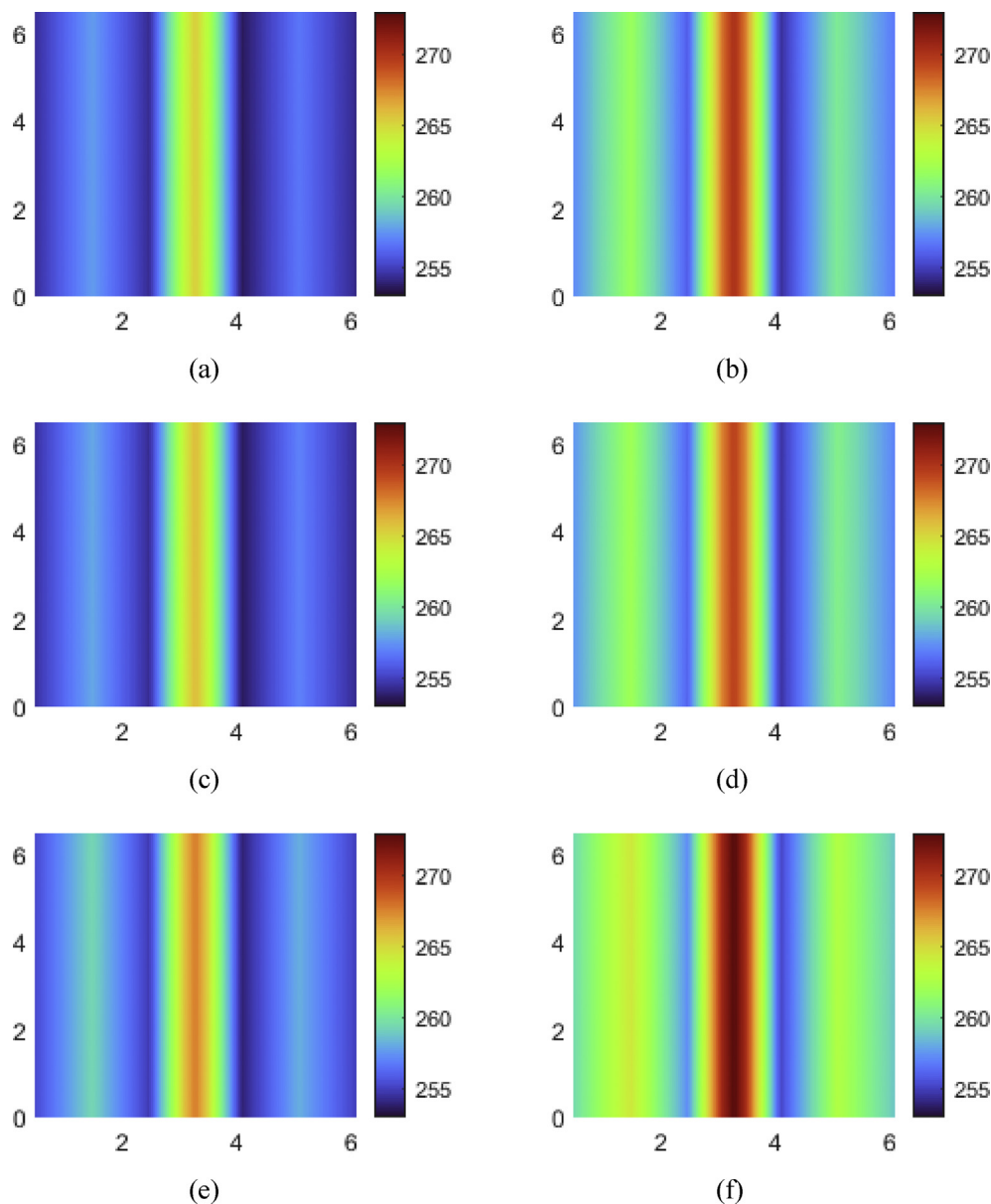


Fig. 10 – The temperature distribution of the fuel cell. (When s is 0.4:(a) Case 6 (c) Case 7 (e) Cases 8; When s is 0.8:(b) Case 6 (d) Case 7 (f) Cases 8).

from Case 4, in which the most heat is produced and the heat absorbed by the MEM ranks fourth among the five cases. In Case 8, the current goes up significantly at the initial stage of start-up because ice volume is 0, leading to a sharp increase in the temperature rise speed of the fuel cell. Then, the fuzzy rules stabilize the current to prevent the rapid growth of the ice volume caused by excessive current. Subsequently, the current gradually increases as the ice volume changes. However, the current increase is significantly smaller than that in Case 5 because the heat absorbed by the MEM is always at a high rate. This indicates that the proposed mode balances heat generation and ice volume.

Fig. 10 shows the temperature distribution of each layer of the fuel cell when the ice volume fraction is 0.4 and 0.8 (the maximum ice volume fraction in Case 8). It can be seen from Fig. 10 (a), (c) and (e) that the temperature in Case 8 is the highest. The MEM temperature is 1.8 K and 2.4 K higher than that in Cases 6 and 7 respectively, and the GDL temperature is 2.2 K and 1.6 K higher than that in Cases 6 and 7 respectively. When the ice volume fraction rises to 0.8, the temperature in Case 8 is still the highest, and the MEM as well as the GDL temperature is 2.7 K–3.5 K higher than that in Cases 6 and 7. The simulation results indicate that the ice volume fraction is the same, the temperature of the fuel cell in Case 8 is the highest. The performance of this thermal behavior intuitively demonstrates that the proposed mode facilitates the cold start. Although the temperatures in Case 6 are still lower than in Case 7 for other structures, the MEM and GDL temperatures are higher than in Case 7 by 0.7 K and 0.4 K, respectively, which are consistent with the MEM heat absorption curves in the two cases in Fig. 9. This presents the requirement for a successful cold start from another perspective: When the heat exchange between the MEM and other structures is the same, if more heat is absorbed by the MEM itself, then a higher heat absorption rate for the MEM is achieved, thus increasing the possibility of achieving a cold start. This conclusion is the same as that in Fig. 7: If the heat absorbed by the MEM is the same, the external heat exchange should be reduced.

The data analysis above demonstrates that the proposed mode not only improves the cold start ability of the fuel cell, but also makes the current change constantly. Considering that the performance degradation caused by a cold start is mainly due to the short-term lack of hydrogen and oxygen during current loading [39], and a current density higher than 500 A/m^2 during load cycle causes more performance loss [40], this paper adds the inertia link $1/(T s + 1)$ to the controller to adjust the amplitude and rate of change of the current density during cold start. As shown in Fig. 11, the current amplitude of $I_{T=10}$ is 7.98% less than that of $I_{T=2}$, and the maximum rate of current loading is reduced by 57.44%. Under the condition that the integral value of the current over time remains unchanged, the maximum ice volume and cold start time remain unchanged. This shows that the addition of the inertia link has little effect on the thermal behavior of the fuel cell cold start, and does not reduce cold-start ability. Therefore, the peak shifting and valley filling effect for current, that is, limiting a high current and increasing a low current, can avoid the short-term lack of hydrogen and oxygen during current loading.

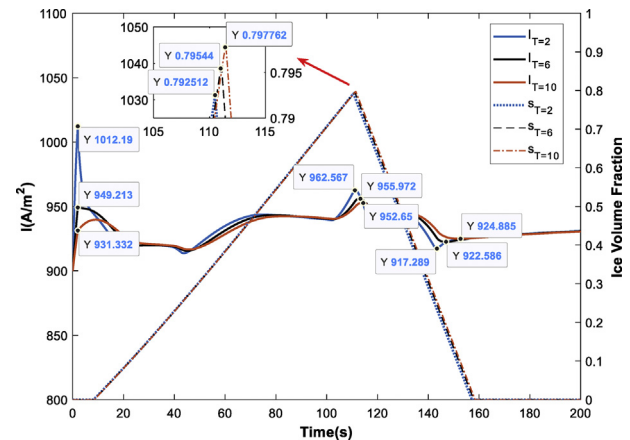


Fig. 11 – Variation of current and ice volume with different time constants.

Conclusion

In this research, a mathematical model is established to simulate the cold start process of the fuel cell, including the calculation of the heat generation of the fuel cell and the mutual heat transfer of its internal structure. Then, we designed a cold start mode in which the ice volume and heat absorbed by the MEM are the fuzzy control input, and current density is the output. This mode is compared with cold-start modes involving constant current and linear current change. The main findings of this study are as follows:

- (1) The proposed mode can control the current, so that the heat exchange in the early stage is reduced, and the MEM heat absorption rate in the later stage is maintained. As indicated by the temperature distribution diagram, the heat absorbed by the MEM in the proposed mode is approximately 20%–33% higher than that in other modes with the same ice volume fraction ($s = 0.8$) at 253 K; furthermore, the outer layers have the lowest temperature in the proposed mode with the same heat absorbed by the MEM (282 J) at 263 K.
- (2) At different ambient temperatures, the proposed mode has the best cold start performance. Compared with other modes at 263 K, the cold startup time is shortened by 25.6s–41.6s and the ice volume fraction is reduced by 29.4–31.8%. Furthermore, only the proposed mode achieves a successful cold start at 253 K.
- (3) Under the same electric quantity, the mode in which the current gradually increases facilitates the cold start. At 263 K, the start mode from low to high current achieves a successful cold start at 210s, whereas the start mode from high to low current fails.
- (4) The inertia link in the controller can adjust the amplitude and rate of change of the current density during cold start. The simulation results show that a larger time constant can avoid the short-term lack of hydrogen and oxygen during current loading: The current amplitude of $I_{T=10}$ is 7.98% less than that of $I_{T=2}$,

and the maximum rate of current loading is 57.44% slower.

Although it has been confirmed that the proposed mode can improve the performance of fuel cell cold start, the specific impact of this strategy on the fuel cell stack should be further studied because of the differences between the fuel cell stack and a single cell. The proposed fuel cell cold-start mode considering the relationship between ice and heat is helpful to the application of fuel cell vehicles.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research is supported by the National Key Research and Development Program of China (Grant No.2017YFB0102800), the Natural Science Foundation of China (Grant No.51775229), and Jilin Province Science and Technology Development Program (Grant No. 20200501010GX, 20190302023GX).

Nomenclature

a_m	water activity
A	electrode area (m^2)
BP	bipolar plate
c	molar concentration of species k ($mol\ m^{-3}$)
C	specific heat ($J\ kg^{-1}\ K^{-1}$)
CL	catalyst layer
D	mass diffusivity (m^2s^{-1})
EW	equivalent weight of dry membrane ($kg\ mol^{-1}$)
F	Faraday's constant ($96,487C\ mol^{-1}$)
GDL	gas diffusion layer
h	heat transfer coefficient, $W\ m^{-2}\ K^{-1}$
h_{fusion}	heat of fusion, $J\ kg^{-1}$
I	current density ($A\ cm^{-2}$)
j	transfer current density ($A\ m^{-3}$)
k	thermal conductivity ($W\ m^{-1}\ K^{-1}$)
K	Permeability(m^2)
m	mass(kg)
$\dot{m}$	mass flow rate, $kg\ s^{-1}$
M	molecular weight ($kg\ mol^{-1}$)
MEA	membrane electrolyte assembly
n_d	electro-osmotic coefficient (H_2O/OH^-)
$\dot{n}_{H_2O}$	water generation/removal rate, $mol\ s^{-1}$
$\dot{n}$	flux of the liquid water ($mol\ m^{-2}\ s^{-1}$)
p	pressure (Pa)
$\dot{Q}$	heat transfer rate(W)
R	universal gas constant, $8.134\ J\ mol^{-1}K^{-1}$
r	resistivity($\Omega\ m$)
s	volume fraction
t	time (s)
T	temperature (K)

V	electrical potential,(V)
v	volume(m^3)

Greek symbols

α	transfer coefficient
δ	thickness (m)
ε	porosity
ζ	water transfer rate(s^{-1})
θ	Contact angle($^\circ$)
λ	membrane water content
ρ	density ($kg\ m^{-3}$)
σ	Surface tension, $N\ m^{-1}$
ξ	stoichiometry ratio

Superscripts and subscripts

0	initial value
a	anode
acc	accumulation
act	activation
bp	bipolar plate
c	cathode, capillary
cl	catalyst layer
clant	Coolant
conc	concentration
dry	dry
D	diffusion
eff	effective value
elec	electrolyte
gas	gas layer
gdl	gas diffusion layer
gen	generate
H_2	Hydrogen
ice	ice
in	inlet
irrev	irreversible
latent	latent
loss	loss
lq	liquid water
mea	membrane electrolyte assembly
net	net
Nernst	Nernst
ohmic	ohmic
out	outlet
O_2	Oxygen
proton	proton
ref	reference value
rev	reversible
sat	saturation
sg	sublimation/desublimation

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